The Q-Schützenberger Identity

Robin Langer

February 10, 2026

Abstract

We define the Q-Schützenberger involution ψ_Q on the pair set $\mathcal{P}_{a,b,k} = \mathcal{G}_{a+k,c} \times \mathcal{G}_{b+k,c}$ as the composition of Q-Burge^{-1} (Burge-glue on the first component) with Q-RSK (RSK-peel on the second component). This is a sign-preserving, weight-preserving map that complements the sign-reversing involution ι of [5]. Together, ι and ψ_Q generate a dihedral group action on the pair set, giving the *Q-Schützenberger identity*: the fixed-point structure of the combined action constrains the positivity of the Q-kernel $\Phi_{a,b}(t)$.

Contents

1 The classical Schützenberger involution **2**
1.1 Two bijections, one involution 2

2 The Q-Schützenberger on pairs **2**
2.1 Ingredients 3
2.2 The composition 3

3 Properties of ψ_Q **3**
3.1 The involution conjecture 4

4 The sign problem: why ψ_Q does not prove positivity **4**
4.1 The fundamental distinction 4
4.2 The missing ingredient: the canonical peel 5
4.3 Classical analogy 5
4.4 The coordinate-independence conjugacy 5

5 The dihedral action **5**
5.1 Orbits and fixed points 6
5.2 The role of ψ_Q 7

6 The Q-Schützenberger identity **7**
6.1 The identity at $b = 0$ 8

7 Special cases **8**
7.1 The case $d = 1$ 8
7.2 The case $d = 2$ (Rogers–Ramanujan) 8
7.3 The case $d = 4$, $c = (2, 2, 0)$ 8

8 Relation to the shifted factorisation **9**

9 The Q-Schützenberger identity: summary **9**

10 The orthogonal structure of the problem **10**

11 Status of the programme **10**
11.1 What is established 10
11.2 What remains open 11
11.3 Where to look for the canonical peel 11

12 Open questions **11**

1 The classical Schützenberger involution

We briefly recall the standard theory to fix notation and motivate the Q-analogue.

1.1 Two bijections, one involution

Both the Burge and RSK local rules give bijections between cylindric plane partitions and pairs (interlacing partition, ALCD):

$$\psi_B \colon \mathrm{CPP}(\pi) \xrightarrow{\sim} \mathrm{IP} \times \mathrm{ALCD}(\pi), \tag{1}$$

$$\psi_R \colon \mathrm{CPP}(\pi) \xrightarrow{\sim} \mathrm{IP} \times \mathrm{ALCD}(\pi). \tag{2}$$

Both are weight-preserving. They differ only in the cascade direction at each face: Burge cascades **up**, RSK cascades **down**.

Definition 1 (Classical Schützenberger involution)**.** The *Schützenberger involution* is the composition:

$$\psi := \psi_R \circ \psi_B^{-1} \colon \mathrm{CPP}(\pi) \to \mathrm{CPP}(\pi). \tag{3}$$

Proposition 2 (Classical properties)**.** *The map ψ satisfies:*

1. $\psi^2 = \mathrm{id}$ (involution),
2. *ψ preserves total weight* ($|\psi(c)| = |c|$),
3. *ψ preserves the base partition γ and the ALCD face labels $\mathfrak{d}$,*
4. *the internal vertex partitions change: $\psi(c) \neq c$ in general.*

The involution ψ is a non-trivial symmetry of $\mathrm{CPP}(\pi)$ that exists purely because there are two valid local rules for the same algebraic identity $T_n P_m = \sum_k P_{m-k} T_{n-k}$.

2 The Q-Schützenberger on pairs

We now construct the Q-analogue on the pair set from [5], Section 10.

2.1 Ingredients

Recall from [5], Sections 12–13:

- **Q-Burge** $\mathfrak{B}$: Burge-peel on the first component. Maps $((S, \mathfrak{d}),\ (S', \mathfrak{d}')) \mapsto ((S \cup \{\ell_B\}, \mathfrak{d}_B),\ (S', \mathfrak{d}'))$. Sign-reversing (changes $|S|$ by $+1$).
- **Q-RSK** $\mathfrak{R}$: RSK-peel on the second component. Maps $((S, \mathfrak{d}),\ (S', \mathfrak{d}')) \mapsto ((S, \mathfrak{d}),\ (S' \cup \{\ell_R\}, \mathfrak{d}'_R))$. Sign-reversing (changes $|S'|$ by $+1$).
- **Q-Burge**$^{-1} = \mathfrak{B}^{-1}$: Burge-glue on the first component. Given $i \in S$, removes i from S and inserts into $\mathfrak{d}$ via Burge forward (cascade **up**). Sign-reversing (changes $|S|$ by -1).
- **Q-RSK**$^{-1} = \mathfrak{R}^{-1}$: RSK-glue on the second component. Removes $i' \in S'$ and inserts into $\mathfrak{d}'$ via RSK forward (cascade **down**). Sign-reversing.

2.2 The composition

Definition 3 (Q-Schützenberger)**.** The *Q-Schützenberger map* is the composition:

$$\boxed{\psi_Q := \mathfrak{R} \circ \mathfrak{B}^{-1} \colon \mathcal{P}_{a,b,k} \to \mathcal{P}_{a,b,k}.} \tag{4}$$

Concretely, given a pair $p = ((S, \mathfrak{d}),\ (S', \mathfrak{d}'))$ at cascade level k:

1. **Burge-glue the first component.** Select the canonical element $i \in S$ (determined by the Burge forward rule) and apply Burge-glue:

$$(S, \mathfrak{d}) \xrightarrow{\mathfrak{B}^{-1}} (S \setminus \{i\},\ \tilde{\mathfrak{d}}).$$

   The element i is absorbed into the ALCD via the Burge cascade (**up**).

2. **RSK-peel the second component.** Apply RSK-peel to the (unchanged) second component:

$$(S', \mathfrak{d}') \xrightarrow{\mathfrak{R}} (S' \cup \{\ell_R\},\ \mathfrak{d}'_R).$$

   The element ℓ_R is extracted from the ALCD via the RSK cascade (**up**).

The result is:

$$\psi_Q(p) = \big((S \setminus \{i\},\ \tilde{\mathfrak{d}}),\ (S' \cup \{\ell_R\},\ \mathfrak{d}'_R)\big). \tag{5}$$

Remark 4 (Both steps cascade **up**)**.** The Burge-glue uses Burge forward (cascade **up**) and the RSK-peel uses RSK inverse (cascade **up**). Both steps push data in the same direction. This is not a coincidence: it reflects the fact that $\mathfrak{B}^{-1}$ inverts the **down** cascade of Burge^{-1} by going **up**, and $\mathfrak{R}$ inverts the **down** cascade of RSK forward by going **up** via RSK^{-1}.

3 Properties of ψ_Q

Proposition 5 (Sign and weight preservation)**.** *The map ψ_Q satisfies:*

1. ***Sign-preserving:*** $(-1)^{|S(\psi_Q(p))|+|S'(\psi_Q(p))|} = (-1)^{|S(p)|+|S'(p)|}$.
2. ***Weight-preserving:*** $\mathrm{wt}(\psi_Q(p)) = \mathrm{wt}(p)$, *where* $\mathrm{wt}(p) = \sigma(S) + |\mathfrak{d}| + \sigma(S') + |\mathfrak{d}'|$.

3. ***Cascade-preserving:*** *ψ_Q maps $\mathcal{P}_{a,b,k}$ to $\mathcal{P}_{a,b,k}$ (both components remain at the same ground-set level).*

Proof. (1) The Burge-glue step changes $|S|$ by -1; the RSK-peel step changes $|S'|$ by $+1$. Net change in $|S| + |S'|$: $-1 + 1 = 0$.

(2) Burge-glue preserves the first-component weight ($\sigma(S \setminus \{i\}) + |\tilde{\mathfrak{d}}| = \sigma(S) + |\mathfrak{d}|$), and RSK-peel preserves the second-component weight.

(3) Burge-glue maps $(S, \mathfrak{d}) \in \mathcal{G}_{a+k,c}$ to $(S \setminus \{i\}, \tilde{\mathfrak{d}}) \in \mathcal{G}_{a+k,c}$ (the ground-set level $a + k$ is unchanged: the subset shrinks by one element, the ALCD grows by one level, and $(a+k)-(|S|-1) = (a + k - |S|) + 1$). Similarly for the second component. □

3.1 The involution conjecture

Conjecture 6 (Q-Schützenberger involution)**.** *$\psi_Q^2 = \mathrm{id}$ on $\mathcal{P}_{a,b,k}$.*

In the classical case, $\psi^2 = \mathrm{id}$ follows from the Bender–Knuth involution relations: the transposition of two adjacent rows of the growth diagram, implemented via Burge and RSK, is self-inverse. The Q-analogue requires that the composition

$$(\mathfrak{R} \circ \mathfrak{B}^{-1}) \circ (\mathfrak{R} \circ \mathfrak{B}^{-1}) = \mathrm{id},$$

i.e., that Burge-glue followed by RSK-peel, applied twice, returns to the original pair. This amounts to a compatibility condition between the Burge and RSK cascade directions at the ground-set level.

Remark 7 (Sufficient condition)**.** A sufficient condition for $\psi_Q^2 = \mathrm{id}$ is that the element i glued by $\mathfrak{B}^{-1}$ and the element ℓ_R peeled by $\mathfrak{R}$ satisfy a "round-trip" property: applying ψ_Q twice returns i to S and removes ℓ_R from S'. In the classical case, this is guaranteed because the growth-diagram row transposition is self-inverse. At the pair level, the round-trip property needs verification for each profile c.

Remark 8 (The $d = 1$ case)**.** When $d = 1$, $Q_n = 1$ for all n, the ground set $\mathcal{G}_{n,c}$ reduces to the standard ALCD set (with $S = \emptyset$ for the only non-trivial contribution), and ψ_Q reduces to the classical Schützenberger involution. In this case, $\psi_Q^2 = \mathrm{id}$ is the classical theorem.

4 The sign problem: why ψ_Q does not prove positivity

4.1 The fundamental distinction

Proving that $Q_{n,c}(q) \in \mathbb{Z}_{\geq 0}[q]$ requires a **sign-reversing** involution $\iota \colon \mathcal{G}_{n,c} \to \mathcal{G}_{n,c}$ that pairs elements of opposite sign ($|S'| \equiv |S| + 1 \pmod 2$) while preserving weight. The Q-Schützenberger map ψ_Q is **sign-preserving** (Proposition 5): it fixes the total $|S| + |S'|$, and therefore cannot perform the cancellations that establish positivity.

Map	Sign	Direction	Proves positivity?
ψ_Q (Q-Schützenberger)	preserving	horizontal	No
ι (peel/glue involution)	reversing	vertical	Yes (if canonical peel exists)

The directions are:

- **Horizontal** (ψ_Q): moves between the Burge and RSK realisations at the *same* $|S| + |S'|$. This is a change of coordinates—column cascade vs. row cascade—that does not alter the subsets.

- **Vertical** (ι): moves between subset levels $|S|+|S'|$ and $|S|+|S'|\pm 1$, trading subset elements for cylindric structure. This is the sign-changing operation that drives cancellation.

4.2 The missing ingredient: the canonical peel

The sign-reversing involution ι requires the **canonical peel choice rule**: given $(S, \mathfrak{d})$, which element k to peel or glue, and whether to peel or glue ([5], Conjecture 8.1). The framework of Sections 2–3 guarantees that the type signature and weight condition are satisfied for *any* valid choice of k. But the choice must be deterministic, involutive, and must produce fixed points only at even $|S| + |S'|$. The Q-Schützenberger map ψ_Q does not address this choice.

4.3 Classical analogy

The situation mirrors the classical theory.

Classical	Q**-enriched**
Schützenberger involution on SYT	Q-Schützenberger ψ_Q on pairs
Symmetry (transposes P- and Q-tableaux)	Symmetry (swaps Burge/RSK coordinates)
Does not prove Littlewood–Richardson rule	Does not prove $Q_{n,c} \geq 0$
Jeu de taquin / crystal operators prove LR	Canonical peel rule proves positivity
Franklin's sign-reversing involution	Sign-reversing ι for positivity

Just as the Schützenberger involution on standard Young tableaux is a deep symmetry that does not by itself establish the Littlewood–Richardson rule, the Q-Schützenberger is a structural result about the *duality* of the pair-level framework, not a *cancellation* result.

4.4 The coordinate-independence conjugacy

Despite not proving the conjecture, ψ_Q constrains the search for the canonical peel.

Proposition 9 (Conjugacy constraint)**.** *If the canonical peel rule is* coordinate-independent *— depending only on partition sizes and abstract structure, not on the column-vs-row resolution of overlaps—then the Burge and RSK peel/glue involutions are conjugate via ψ_Q:*

$$\mathfrak{V}^R = \psi_Q \circ \mathfrak{V}^B \circ \psi_Q^{-1}. \tag{6}$$

Proof. If the canonical choice depends only on $(S, \mathfrak{d})$ at the abstract level (not on which coordinate system resolves overlaps), then $\mathfrak{V}^B$ and $\mathfrak{V}^R$ make the same peel/glue decisions. They differ only in the face-level geometry (Burge vs. RSK cascade), which is exactly the transformation performed by ψ_Q. □

Corollary 10. *Under the hypothesis of Proposition 9, finding the canonical peel for Burge automatically gives the RSK version. The conjugacy* (6) *also implies $\psi_Q\colon \mathrm{Fix}(\mathfrak{V}^B) \xrightarrow{\sim} \mathrm{Fix}(\mathfrak{V}^R)$: the two "charge decompositions" are equivalent.*

5 The dihedral action

The pair set $\mathcal{P}_{a,b,k}$ now carries two distinguished maps:

Map	Definition	Sign	Order
ι	Q-Burge–RSK involution ([5], Def. 13.1)	reversing	2
ψ_Q	$\mathfrak{R} \circ \mathfrak{B}^{-1}$ (Definition 3)	preserving	2 (conj.)

Proposition 11 (Group structure)**.** *Assuming Conjecture 6, the group $\Gamma = \langle \iota, \psi_Q \rangle$ generated by the two involutions is a dihedral group:*

$$\Gamma \cong D_m \quad \textit{for some } m \geq 1,$$

where m is the order of $\iota \circ \psi_Q$. The group acts on $\mathcal{P}_{a,b,k}$ preserving weight and with the sign character:

$$\operatorname{sgn}(\gamma) = \begin{cases} +1 & \textit{if } \gamma \textit{ is a product of an even number of } \iota\textit{'s}, \\ -1 & \textit{if odd.} \end{cases}$$

Proof. The group generated by two involutions ι and ψ_Q is always dihedral (or cyclic if they commute). Since ι is sign-reversing and ψ_Q is sign-preserving, the sign character distinguishes the two cosets of the rotation subgroup $\langle \iota \circ \psi_Q \rangle$ in Γ. □

5.1 Orbits and fixed points

The Γ-orbits on $\mathcal{P}_{a,b,k}$ decompose into three types:

Definition 12 (Orbit types)**.** Let $\mathcal{O}$ be a Γ-orbit in $\mathcal{P}_{a,b,k}$.

1. **Type I (cancelled).** $\mathcal{O}$ contains elements of both signs. These contribute zero to $Q_{a+k}Q_{b+k}$ after signed summation.
2. **Type II (positive).** $\mathcal{O}$ consists entirely of elements with sign $+1$ (i.e., $|S| + |S'|$ even for all elements). Every element is a fixed point of ι (since ι is sign-reversing, it must fix elements that have no opposite-sign partner in $\mathcal{O}$).
3. **Type III (negative).** $\mathcal{O}$ consists entirely of elements with sign -1. Same reasoning: every element is a fixed point of ι.

Proposition 13 (Positivity criterion)**.** *$Q_{a+k} \cdot Q_{b+k} \in \mathbb{Z}_{\geq 0}[q]$ if and only if there are no Type III orbits at cascade level k. Equivalently: every fixed point of ι in $\mathcal{P}_{a,b,k}$ has sign $+1$.*

Proof. The signed generating function is:

$$\begin{aligned} Q_{a+k} \cdot Q_{b+k} &= \sum_{p \in \mathcal{P}_{a,b,k}} (-1)^{|S(p)|+|S'(p)|}\, q^{\mathrm{wt}(p)} \\ &= \sum_{p \in \mathrm{Fix}(\iota)} (-1)^{|S(p)|+|S'(p)|}\, q^{\mathrm{wt}(p)}, \end{aligned}$$

since the non-fixed-point pairs cancel in opposite-sign pairs under ι. This is non-negative if and only if every fixed point has $|S(p)| + |S'(p)|$ even. □

5.2 The role of ψ_Q

The Q-Schützenberger ψ_Q acts *within* the fixed-point set $\mathrm{Fix}(\iota)$ (since ψ_Q preserves sign, it maps fixed points of ι to fixed points of ι, provided ι and ψ_Q commute on $\mathrm{Fix}(\iota)$). This gives additional structure:

Proposition 14 (Refinement of fixed points)**.** *If ψ_Q restricts to an involution on $\mathrm{Fix}(\iota)$, then the fixed-point contribution decomposes as:*

$$\sum_{p\in\mathrm{Fix}(\iota)} (-1)^{|S|+|S'|}\, q^{\mathrm{wt}(p)} \;=\; \underbrace{\sum_{p\in\mathrm{Fix}(\iota)\cap\mathrm{Fix}(\psi_Q)} (-1)^{|S|+|S'|}\, q^{\mathrm{wt}(p)}}_{\textit{doubly fixed}} \;+\; \underbrace{\sum_{\substack{p\in\mathrm{Fix}(\iota)\\ \psi_Q(p)\neq p}} (-1)^{|S|+|S'|}\, q^{\mathrm{wt}(p)}}_{\textit{ψ_Q-paired, same sign}}.$$

Since ψ_Q is sign-preserving, the ψ_Q-paired terms on the right contribute positively *(they have the same sign and pair within the same sign class). The doubly fixed points are the irreducible core.*

6 The Q-Schützenberger identity

We now state the main identity.

Theorem 15 (Q-Schützenberger identity)**.** *Let $c = (c_0, c_1, c_2)$ be a rank-3 profile with $d = c_0 + c_1 + c_2$, and let $a, b, k \geq 0$. The Q-kernel coefficient $Q_{a+k}(q) \cdot Q_{b+k}(q)$ admits the decomposition:*

$$\boxed{Q_{a+k}\, Q_{b+k} = \sum_{p\in\mathrm{Fix}(\iota)\cap\mathrm{Fix}(\psi_Q)} q^{\mathrm{wt}(p)} \;+\; 2 \sum_{\substack{\{p,\psi_Q(p)\}\subseteq\mathrm{Fix}(\iota)\\ \psi_Q(p)\neq p}} q^{\mathrm{wt}(p)},} \tag{7}$$

where the second sum runs over unordered ψ_Q-orbits of size 2 within $\mathrm{Fix}(\iota)$, and both sums contribute non-negative coefficients (all terms have $|S(p)| + |S'(p)|$ even, hence sign $+1$).

Proof. By Proposition 13, the signed sum reduces to $\mathrm{Fix}(\iota)$. By Proposition 14, this decomposes into doubly-fixed points and ψ_Q-pairs within $\mathrm{Fix}(\iota)$. Since ψ_Q is weight-preserving and sign-preserving, each ψ_Q-pair $\{p, \psi_Q(p)\}$ contributes $2\, q^{\mathrm{wt}(p)}$ (both elements have the same weight and sign $+1$). The doubly-fixed points contribute $1 \cdot q^{\mathrm{wt}(p)}$ each. □

Remark 16 (Conditional on two conjectures)**.** Theorem 15 is conditional on:

(a) The existence of the sign-reversing involution ι with only positive fixed points (Warnaar's Conjecture 2.7 at the pair level), and

(b) Conjecture 6 ($\psi_Q^2 = \mathrm{id}$).

Point (a) is the hard conjecture. Point (b) is a structural property of the Burge–RSK interaction at the ground-set level, which should follow from the classical Bender–Knuth theory adapted to the Q-enriched setting.

6.1 The identity at $b = 0$

Setting $b = 0$ in (7) gives a formula for Q_a itself, since $Q_0 = 1$:

$$Q_{a+k} = Q_{a+k} \cdot Q_k\big|_{b=0} = Q_{a+k} \cdot 1. \tag{8}$$

More precisely, the pair set at $b = 0$ is $\mathcal{G}_{a+k} \times \mathcal{G}_k$, and $\mathcal{G}_0 = \{(\emptyset, \mathfrak{d}_0)\}$ is a singleton. The second component is trivially fixed by both ι and ψ_Q, so:

$$Q_a = [t^0]\,\Phi_{a,0}(t) = \sum_{\substack{p \in \mathrm{Fix}(\iota) \cap \mathrm{Fix}(\psi_Q) \\ p \in \mathcal{G}_a \times \mathcal{G}_0}} q^{\mathrm{wt}(p)} \;+\; 2 \sum_{\{p, \psi_Q(p)\} \subseteq \mathrm{Fix}(\iota) \cap (\mathcal{G}_a \times \mathcal{G}_0)} q^{\mathrm{wt}(p)}. \tag{9}$$

Since the second component is the singleton $(\emptyset, \mathfrak{d}_0)$, the pair reduces to a single ground-set element $(S, \mathfrak{d}) \in \mathcal{G}_{a,c}$, and the Q-Schützenberger identity becomes a statement about the ground set alone:

$$Q_a = \sum_{(S,\mathfrak{d}) \in \mathrm{Fix}(\iota_a)} (-1)^{|S|}\, q^{\sigma(S)+|\mathfrak{d}|} \;\in\; \mathbb{Z}_{\geq 0}[q], \tag{10}$$

where ι_a is the restriction of ι to $\mathcal{G}_{a,c} \times \{(\emptyset, \mathfrak{d}_0)\} \cong \mathcal{G}_{a,c}$.

7 Special cases

7.1 The case $d = 1$

When $d = 1$ (one box per period), $Q_n = 1$ for all n, the ground set $\mathcal{G}_{n,c}$ has a single element $(\emptyset, \mathfrak{d})$ for each ALCD $\mathfrak{d}$ (the Euler-factor subsets $S \neq \emptyset$ contribute cancelling pairs that sum to zero). The Q-Schützenberger ψ_Q reduces to the classical Schützenberger involution, and the identity (7) becomes:

$$Q_{a+k} \cdot Q_{b+k} = 1 = |\,\mathrm{ALCD}_{a+k}(c) \times \mathrm{ALCD}_{b+k}(c)|\big|_{\mathrm{signed}}.$$

7.2 The case $d = 2$ (Rogers–Ramanujan)

For $d = 2$ with profile $c = (2, 0, 0)$: $Q_n = q^{n^2}$ (first Rogers–Ramanujan identity). The product is:

$$Q_{a+k} \cdot Q_{b+k} = q^{(a+k)^2+(b+k)^2}.$$

This is a single monomial, so the pair set at each cascade level has exactly one doubly-fixed point contributing $q^{(a+k)^2+(b+k)^2}$, and all other elements cancel under ι. The Q-Schützenberger acts trivially on the unique fixed point ($\psi_Q = \mathrm{id}$ on $\mathrm{Fix}(\iota)$).

For $c = (1, 1, 0)$: $Q_n = q^{n(n+1)}$ (second Rogers–Ramanujan), and the same monomial structure holds:

$$Q_{a+k} \cdot Q_{b+k} = q^{(a+k)(a+k+1)+(b+k)(b+k+1)}.$$

7.3 The case $d = 4$, $c = (2, 2, 0)$

Here $Q_0 = 1$, $Q_1 = q^2 + q^3 + q^4 + q^5$. The product $Q_1 \cdot Q_1 = (q^2 + q^3 + q^4 + q^5)^2$ has multiple terms, so the doubly-fixed points and ψ_Q-pairs of $\mathrm{Fix}(\iota)$ contribute a non-trivial decomposition. The ψ_Q-orbits of size 2 within $\mathrm{Fix}(\iota)$ account for the "extra factor of 2" in the coefficients of $Q_1^2 = q^4 + 2q^5 + 3q^6 + 4q^7 + 3q^8 + 2q^9 + q^{10}$.

8 Relation to the shifted factorisation

The shifted factorisation ([1], Part III):

$$\Phi_{a,b}(t) = Q_a\, Q_b + t\, \Phi_{a+1,b+1}(t) \tag{11}$$

acquires a refined interpretation through the dihedral action.

At cascade level $k = 0$, the pair set is $\mathcal{P}_{a,b,0} = \mathcal{G}_a \times \mathcal{G}_b$. The Γ-action decomposes it into orbits, and the fixed-point contribution is $Q_a \cdot Q_b$.

At level $k = 1$, the pair set is $\mathcal{G}_{a+1} \times \mathcal{G}_{b+1}$, and the fixed-point contribution is $Q_{a+1} \cdot Q_{b+1}$.

The shifted factorisation says: the $k = 0$ contribution and the $k \geq 1$ contributions assemble into $\Phi_{a,b}(t)$, with each level independently non-negative (assuming the conjecture). The Q-Schützenberger provides an *intra-level symmetry*: within each cascade level k, the fixed points of ι are further organised by ψ_Q into doubly-fixed points and ψ_Q-pairs.

Remark 17 (Inter-level vs. intra-level)**.** The cascade-coupled involution ι can pair elements at *different* cascade levels (Definition 11.2 of [5]). The Q-Schützenberger ψ_Q operates *within* a single level. Together they provide both inter-level cancellation (via ι) and intra-level symmetry (via ψ_Q).

9 The Q-Schützenberger identity: summary

We summarise the logical structure.

Map	Type	Role
ι	sign-reversing involution	cancels opposite-sign pairs
ψ_Q	sign-preserving involution (conj.)	organises fixed points of ι
$\Gamma = \langle \iota, \psi_Q \rangle$	dihedral group	full symmetry of pair set

The identity. The Q-kernel coefficient decomposes as:

$$Q_{a+k}\, Q_{b+k} \;=\; \underbrace{|\operatorname{Fix}(\iota) \cap \operatorname{Fix}(\psi_Q)|_q}_{\text{doubly fixed}} \;+\; \underbrace{2\,|\psi_Q\text{-pairs in } \operatorname{Fix}(\iota)|_q}_{\text{Schützenberger pairs}},$$

where $|\cdot|_q$ denotes the q-weighted count. All terms are non-negative.

The programme.

1. Construct the sign-reversing involution ι (the hard step — Warnaar's conjecture).
2. Verify $\psi_Q^2 = \mathrm{id}$ (the Q-Bender–Knuth relations).
3. Characterise the doubly-fixed points $\operatorname{Fix}(\iota) \cap \operatorname{Fix}(\psi_Q)$ — these are the irreducible positivity core.
4. At $b = 0$: the identity reduces to a formula for Q_a in terms of the ground set $\mathcal{G}_{a,c}$ alone (equation (10)).

10 The orthogonal structure of the problem

The analysis reveals that the positivity problem decomposes into two *orthogonal* components.

	Horizontal (symmetry)	**Vertical (cancellation)**
Map	ψ_Q (Q-Schützenberger)	ι (peel/glue)
Action	Swaps Burge $\leftrightarrow$ RSK	Trades $S \leftrightarrow \mathfrak{d}$
Subsets S, S'	Fixed	Change by ± 1
ALCDs $\mathfrak{d}, \mathfrak{d}'$	Change (coord. system)	Change (peel/glue)
Sign	Preserved	Reversed
Status	**Defined** (this note)	**Conjectural** ([5])
Role	Constrains the problem	Solves the problem

The horizontal component (ψ_Q) tells us the *stage is well-built*: the pair-level framework supports a clean duality between two coordinate systems, confirming that this is the correct level of abstraction for the Q-enriched theory.

The vertical component (ι) is the *actor*: it must perform the sign-reversing cancellation that establishes positivity. Its construction requires the canonical peel rule, which remains open.

The two components interact through the dihedral group $\Gamma = \langle \iota, \psi_Q \rangle$ (Section 5), and through the conjugacy $\mathfrak{V}^R = \psi_Q \circ \mathfrak{V}^B \circ \psi_Q^{-1}$ (Proposition 9). But they serve fundamentally different purposes.

11 Status of the programme

11.1 What is established

1. **Ground set.** $\mathcal{G}_{n,c}$ with signed generating function $Q_{n,c}(q)$ ([8, 4]).

2. **Two local rules.** Burge and RSK peel/glue on pairs, with shared type signature, same face label, same weight condition ([6]).

3. **Q-Schützenberger.** $\psi_Q = \mathfrak{R} \circ \mathfrak{B}^{-1}$ well-defined, weight-preserving, sign-preserving (this note).

4. **Locality principle.** Effective operations in the Q-enriched setting must be local—one face at a time—matching the shifted factorisation $\Phi_{a,b}(t) = Q_a Q_b + t\,\Phi_{a+1,b+1}(t)$ ([5], Theorem 7.1).

5. **Propagation obstruction.** The global Schützenberger involution fails in the Q-enriched setting ([5], Section 4), confirming that global approaches do not work.

6. **Cyclic descent injection.** For $d \not\equiv 0 \pmod 3$, the max-1 case has a valid injection establishing monotonicity of $h_{1,c}(q)$ coefficients ([7], Section 8).

7. **Computational verification.** $Q_{n,c}(q) \in \mathbb{Z}_{\geq 0}[q]$ verified for $d = 1, 2, 4, 5, 7$ (all tested profiles pass).

11.2 What remains open

1. **Canonical peel rule** ([5], Conjecture 8.1). The deterministic choice of which element k to peel or glue, given $(S, \mathfrak{d})$. This is the central open problem.

2. **Q-Bender–Knuth relations.** Conjecture 6: $\psi_Q^2 = \mathrm{id}$. Structural property of Burge–RSK interaction at the ground-set level.

3. **Fixed-point characterisation.** Identify the "irreducible" pairs that are fixed by both ι and ψ_Q. Their cardinality at $b = 0$ must be $Q_{n,c}(1) = \left(\frac{(d+1)(d+2)}{6} - 1\right)^n$.

4. **Charge statistic.** The weight function on fixed points defines the "missing charge." Understanding its structure (analogue of Lascoux–Schützenberger charge?) is a further goal.

5. **The $d \equiv 0 \pmod 3$ case.** The cyclic descent injection fails here (balanced slack has no cyclic descent). A different mechanism is needed, possibly involving the pair-based framework at deeper ALCD levels.

11.3 Where to look for the canonical peel

The Q-Schützenberger, while not proving the conjecture, narrows the search in three ways.

1. Coordinate independence. By Proposition 9, a coordinate-independent peel rule for Burge automatically gives the RSK version. This suggests looking for a rule based on *abstract* properties: sizes of horizontal strips, face-label magnitudes, or diagonal weights—rather than specific column or row indices.

2. The shifted factorisation as guide. The recurrence $\Phi_{a,b}(t) = Q_a Q_b + t\,\Phi_{a+1,b+1}(t)$ peels one layer at a time ([5], Section 7.2). The canonical peel should correspond to the $k = 1 \to k = 0$ reduction: peel the face whose removal corresponds to stripping the leading term from the Q-kernel.

3. The cyclic descent as prototype. For the max-1 case ($d \not\equiv 0 \pmod 3$), the cyclic descent injection ([7], Section 8) provides an explicit peel rule based on the slack vector. This rule is coordinate-independent and produces an injection from weight-$(w-1)$ to weight-w configurations. Extending this to general ALCD depth is the key challenge.

12 Open questions

1. **Q-Bender–Knuth.** Does $\psi_Q^2 = \mathrm{id}$? Is there a Q-analogue of the Bender–Knuth involution that mediates between the Burge and RSK cascades at the ground-set level?

2. **Order of $\iota \circ \psi_Q$.** What is the order m of the dihedral group $\Gamma = \langle \iota, \psi_Q \rangle$? In the classical case, ψ has order 2 and $\iota \circ \psi$ may have larger order. Does the Q-enrichment change the group structure?

3. **Doubly-fixed-point formula.** Is there a closed form for the q-count of $\mathrm{Fix}(\iota) \cap \mathrm{Fix}(\psi_Q)$? In the $d = 2$ case, this is a single monomial $q^{(a+k)^2+(b+k)^2}$ (or $q^{(a+k)(a+k+1)+(b+k)(b+k+1)}$). Is there a quadratic-form structure for general d?

4. **Crystal-base interpretation.** The classical Schützenberger involution has an interpretation via crystal bases for $\widehat{\mathfrak{sl}}_r$ (cf. [3]). Does the Q-Schützenberger correspond to a crystal operator on the Q-enriched pair set?

5. **Computational verification.** For small d and n, enumerate $\mathcal{P}_{a,b,k}$, construct ι and ψ_Q explicitly, verify the identity (7), and tabulate the doubly-fixed-point counts.

References

[1] R. Langer, *The Q-Operator and its Adjoint: From Level 3 Operator Traces to Cascade Commutators*, companion note, 2026.

[2] A. Borodin, *Periodic Schur process and cylindric partitions*, Duke Math. J. **140** (2007), 391–468.

[3] R. Langer, *From Commutation Relations to Growth Diagrams: A Summary of the Operator–Bijection Correspondence*, companion note.

[4] R. Langer, *Towards an Involution Proof of Positivity for $Q_{n,c}(q)$*, companion note.

[5] R. Langer, *The Generalized Local Rule and the Burge–RSK Involution*, companion note, 2026.

[6] R. Langer, *Two Equivalent Local Rules on Ground-Set Pairs*, companion note, 2026.

[7] R. Langer, *Weighted Local Rules and the Cyclic Descent Injection*, companion note.

[8] R. Langer, *The Underlying Set: Cylindric Plane Partitions, ALCDs, and the Ground Set for the Involution*, companion note.

[9] R. Langer, *Cylindric Partitions and Bijective Proofs*, PhD thesis, University of Melbourne.

[10] M. A. A. van Leeuwen, *The Robinson–Schensted and Schützenberger algorithms, an elementary approach*, Electron. J. Combin. **3** (1996), R15.